

Characterization of a high-pressure turbine stage for an industrial power production gas turbine

version: **1.0**
date: **22/02/2023**

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1 Introduction

As a member of the turbine aerodynamic team, you are asked to characterize the aerodynamics of a turbine stage for an industrial F-class gas turbine engine. To do so, You will use the results of the 1D stage design to develop a rotor blade geometry and to use an open source computational tool-chain to characterize the aerodynamics of your blade design.

The project includes the following steps:

1. Perform a 1D mean-line design of a turbine stage given certain input boundary conditions and the target turbine application,
2. Perform the 2D aerodynamics design of a rotor blade passage that performs the velocity triangles that resulted from your 1D mean-line design,
3. Use a computational tool-chain to construct a blade profile, predict the rotor flow through a CFD steady RANS computation and analyze the blade aerodynamics:
 - Use the library Parablade to construct a blade profile (coordinates and blade passage geometry) that implements your 2D blade design achieved at the second project step,
 - Use the GMSH software to mesh the flow domain,
 - Use SU2 software to simulate the 2D steady compressible turbulent flow around the blade,
 - Post-process and analyze the aerodynamic blade performance using Python and Paraview.

2 The 1D mean-line turbine design

2.1 Input data

The aim of the project is to perform the design of the the first HP stage of a four-stage industrial turbine. An example of the architecture of such a gas turbine engine, the

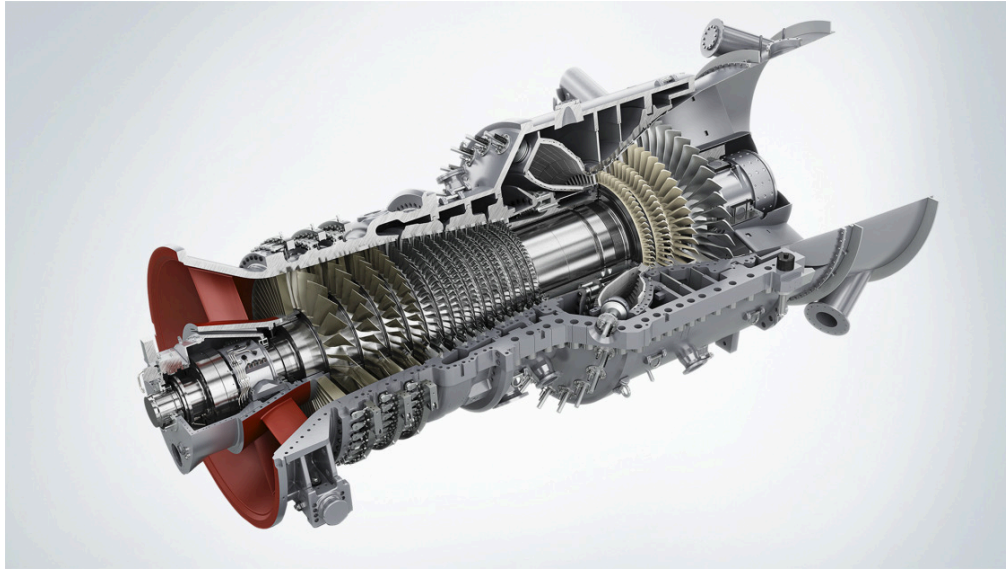


Figure 1: Siemens SGT5-4000F core engine

Siemens SGT5-4000F, is shown in Fig. 1. The 4-stages turbine module is visible on the far right side of the gas turbine.

You design a turbine stage considering the following input data:

- The working fluid is a calorifically perfect diatomic gas with the following properties:
 - the ratio of specific heats $\gamma = 4.0/3.0$,
 - the specific gas constant $R = 287 \text{ J kg}^{-1} \text{ K}^{-1}$.
- Turbine massflow rate $\dot{m} = 750 \text{ kg/s}$, the fuel-air ratio for a heavy duty industrial gas turbine is $far = 0.025$.
- Turbine stage (extracted) power at the shaft $P = 330 \text{ MW}$. Recall that the turbine also drives the compressor which has to be taken into account in the power budget.
- Overall static pressure ratio for the compressor $\pi_C = 20$.
- Compressor admission static conditions $t_{adm} = 15^\circ \text{C}$ and $p_{adm} = 101325 \text{ Pa}$.
- Compressor isentropic efficiency $\eta_{sIC} = 0.92$.
- Mach number in the combustion chamber $M_{comb} = 0.15$.
- Turbine inlet static temperature for an F-class turbine $TIT = 1400^\circ \text{C}$
- Turbine inlet yaw angle $\alpha_1 = 0^\circ$ (fully axial)

$$U = \omega R$$

$$R = \frac{U}{\omega} = \frac{350}{2\pi} \times \frac{3000}{60} = 2745.2 \text{ m}$$

$$\omega = \frac{2\pi}{N} = \frac{2\pi}{3000}$$

$$C_p - C_v = R$$

$$\frac{C_p}{C_v} = \gamma$$

- As the shaft is connected to a synchronous generator the rotational speed is $N_{rot} = 3000 \text{ rpm}$.
- Due to centrifugal stress limitations, the blade peripheral velocity is limited such that at the mean radius $U_m = \omega R_m = 350 \text{ m/s}$.
- In order to ensure a safe and realizable turbine, your design must respect the general mechanical design constraint that AN^2 value must be lower than $3E+07 \text{ m}^2 \text{rpm}^2$.

$$C_p - \frac{C_p}{\gamma} = R$$

$$\left(1 - \frac{1}{\gamma}\right) C_p = R$$

$$C_p = \frac{R}{\left(1 - \frac{1}{\gamma}\right)}$$

In your design process, you must make the following assumptions:

- The **mean-line** radius R_{mean} does not change across the turbine stage, and its value is given as input data,
- The blading efficiencies of both stator and rotor rows are to remain fixed throughout your 1D design at the value of $\eta_{S,R} = 0.90$

2.2 Expected output

You are expected to provide the following results as the outcome of your 1D mean-line design work:

- A description of the design procedure, including the step-by-step calculations used to carry out the 1D mean-line analysis with justification of the design choices and underlying assumptions taken,
- The thermodynamic quantities at the relevant turbine stations (stator inlet, stator outlet / rotor inlet, turbine outlet): total (absolute and relative) and static pressures, temperatures, velocities, flow angles
- a graphical representation of the velocity triangles and of the corresponding H-S diagram
- The annulus sizing at the relevant turbine stations: channel span height and hub and tip radius

3 Rotor blade design

3.1 Input data

For this part of the project, you propose a preliminary design of the rotor blade geometry (rotor airfoil profile geometry and rotor passage geometry). Note: the design and evaluation of the stator row is not requested. The rotor blade design must start from the results of the 1D design (velocity triangles, thermodynamic state and annulus sizing). You define your blade design by selecting:

- the rotor blade aspect ratio $AR = H/C_x$ (where H is the blade span at the rotor inlet and C is the rotor chord)

- the rotor blade pitch-to-chord ratio,
- the stagger angle
- the leading and trailing edge radius
- a suitable blade load distribution style for the given application.

Use the guidelines given in class and good engineering practices dictated by the given turbine application to make reasonable design choices for geometrical parameters not expressly provided as input data. The optimum pitch-to-chord ratio can be determined by means of the Zweifel coefficient defined for incompressible flows. For this project, you must enforce a Zweifel coefficient of 1.0.

3.2 Expected output

You are expected to provide the following results as the outcome of your preliminary blade design:

- A description of the design procedure, including the step-by-step calculations used for the selection of the main blade geometrical parameters, with justification of the design choices and underlying assumptions taken,
- A table with the main blade parameters

4 Rotor blade construction and CFD evaluation

For this part of the project, you are tasked with the construction of a blade profile based on the results of the blade design performed at the previous step. Use the library Parablade to produce an airfoil profile defined by a camber line and a thickness distribution that comply with your target blade load distribution style.

4.1 Expected output

You are expected to provide the following results as the outcome of your blade construction and CFD evaluation:

- A figure of the constructed blade passage geometry,
- Evaluation of the profile curvature and passage area change along the axial direction,
- The CFD setup and settings,
- Isentropic Mach number (or blade static pressure) distribution over the blade profile,
- Commented colormaps of the Mach number distribution, total enthalpy, entropy, pressure.

- A velocity profile highlighting the momentum deficit in the wake.
- The CFD prediction results such as: pitchwise distributions of total pressure, flow angle, Mach number extracted at half-a-chord downstream the rotor trailing edge

5 Project evaluation

The project is evaluated according to the following criteria:

1. Quality of the report (e.g: quality of figures, clarity of writing, organization, etc.)
2. Depth and quality of the justifications provided for the design choices made through the different project steps.
3. Correctness and quality of the design choices.

stage loading $\psi = \frac{\Delta H}{U^2}$

flow coefficient $\phi = \frac{V_{ax}}{U}$
 ? axial velocity $\rightarrow \frac{C_a}{U}$
 blade peripheral speed

Degree of reaction $\lambda = \frac{h_2 - h_x}{h_3 - h_x} = \frac{T_2 - T_1}{T_3 - T_1}$

$\pi_c = \frac{p_{03}}{p_{01}} \Rightarrow p_{03} = 20 \times p_{01}$

$\eta_{TT} = ?$
 $1 - \left[\frac{1}{\psi \left(\frac{V}{a_0} \right)^2} - (r-1) \right] M_{ax}^2 \cdot \xi_{SR} ?$



$$\frac{T_0}{T} = 1 + \frac{\gamma-1}{2} M^2$$

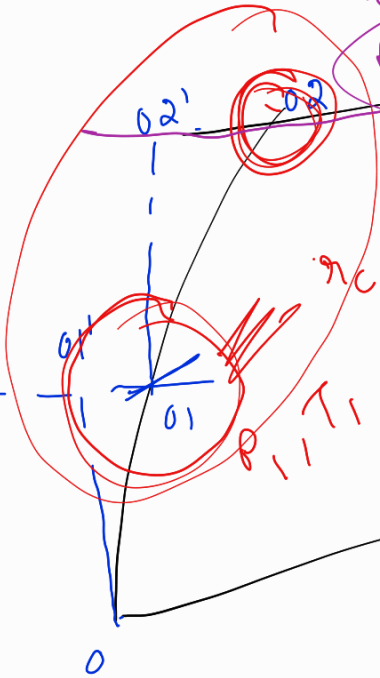
$$\frac{p_0}{p} = \left(\frac{T_0}{T} \right)^{\frac{\gamma}{\gamma-1}} = \left(1 + \frac{\gamma-1}{2} M^2 \right)^{\frac{\gamma}{\gamma-1}}$$

$T \rightarrow$ static
 $T_0 \rightarrow$ total
 $T_{0s} \rightarrow$ isentropic

$$M = \frac{c}{a}$$

$$a = \sqrt{\gamma R T}$$

$$\eta_{isc} = \frac{T_{02s} - T_1}{T_{02} - T_1}$$



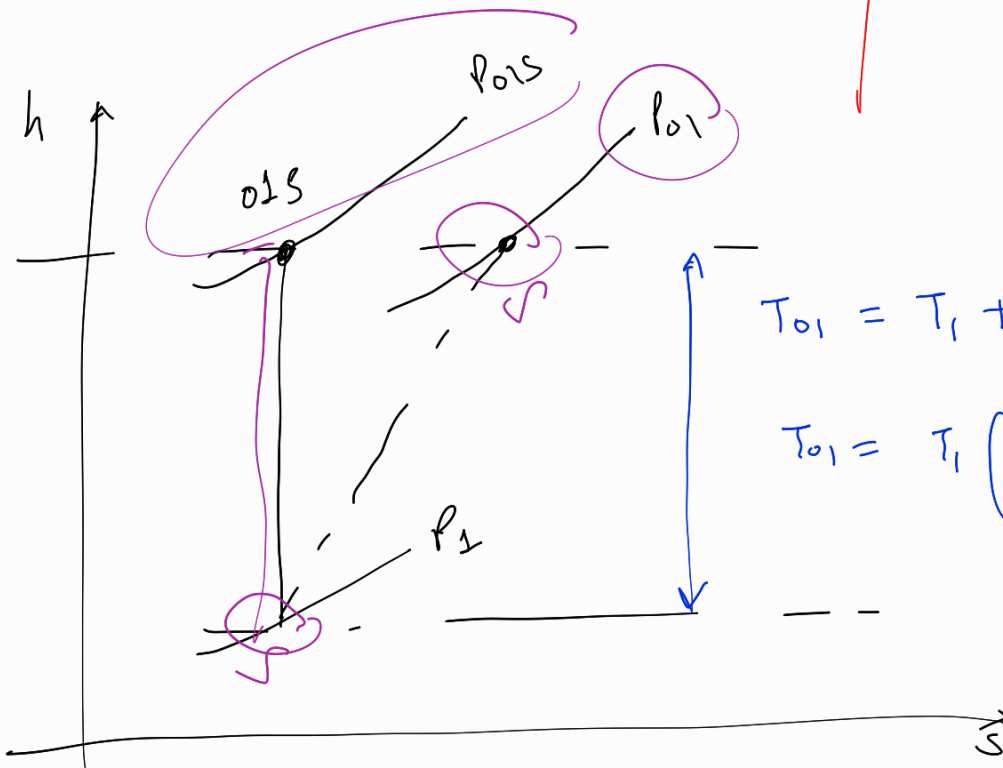
$p_1 \checkmark$

$T_1 \checkmark$

$T_3 \checkmark$

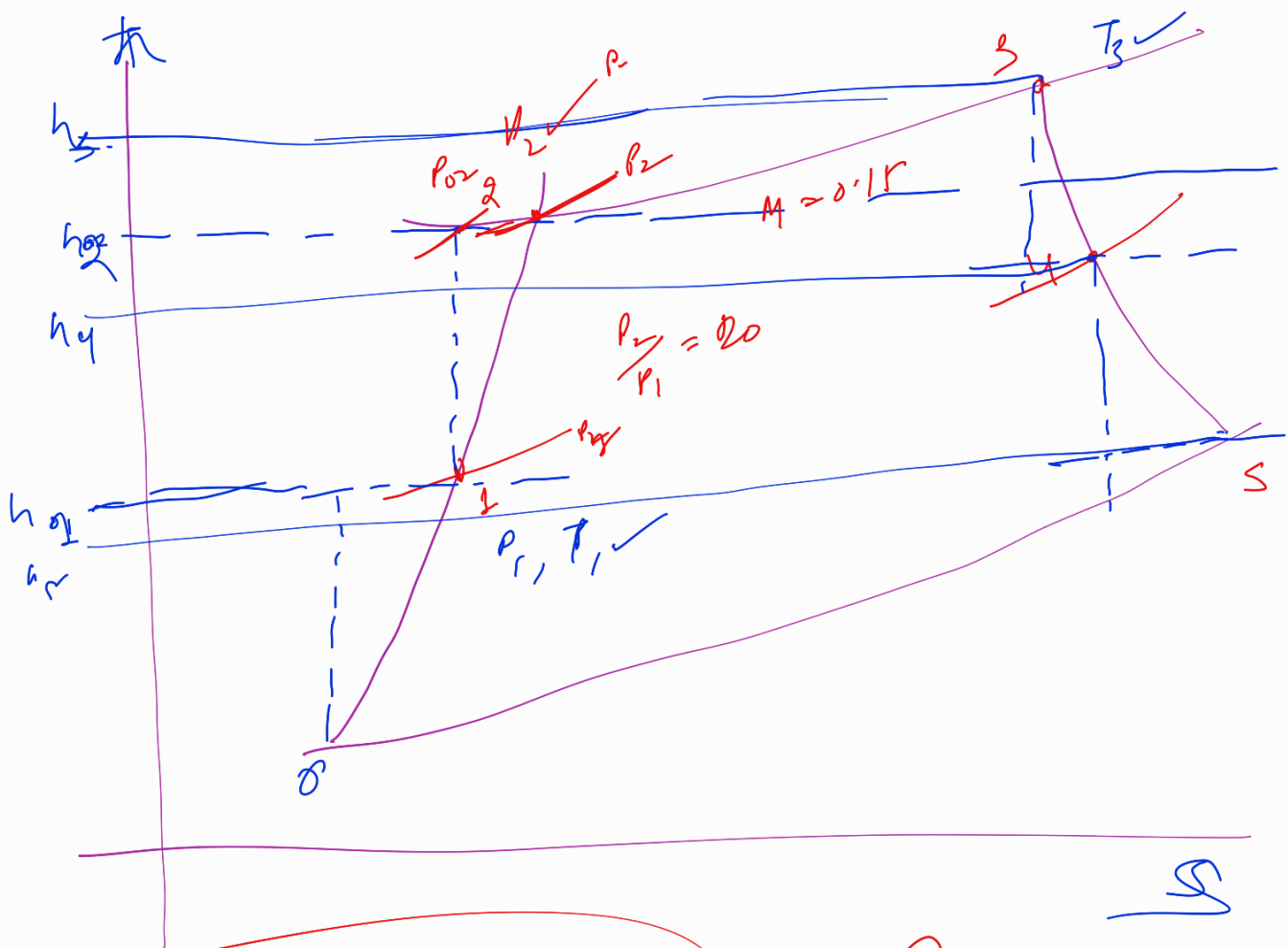
$$\frac{p_2}{p_1} = 20 \rightarrow p_2 \checkmark$$

$$\frac{p_{01}}{p_3} = \frac{T_{01}}{T_3}$$



$$T_{01} = T_1 + \frac{c^2}{2C_p}$$

$$T_{01} = T_1 \left(1 + \frac{\gamma-1}{2} M^2 \right)$$



$$C_p = \frac{\gamma R}{(\gamma - 1)}$$

$$\begin{cases} C_p - C_v = R \\ \frac{C_p}{C_v} = \gamma \end{cases}$$

$$\Delta n_T < \Delta n_{T, \text{real}} \rightarrow p_3' < p_3$$

$$\Delta n_T > \Delta n_{T, \text{real}} \rightarrow p_3' > p_3$$

$$\Delta n_T = \Delta n_{T, \text{real}}$$